

Revision F Electrical Design

Updated 7 October 2026. This revision replaces the relay-based Revision E circuit with a serviceable DIY design. It retains the Arduino Nano Every, three Serial Wombat PCB0046 HSD V2 boards, proportional valve, command and feedback modules, and local supply capacitors. There is no relay, relay regulator, permanent flow meter, or external temperature sensor.

Status and scope

This is a documented bench prototype, not physically tested hardware. No flashable firmware is included. The schematic, connection register and automated checks establish documented connections; they do not establish actual valve behavior, thermal performance or software recovery. Use the same component revisions and check the following before machine installation.

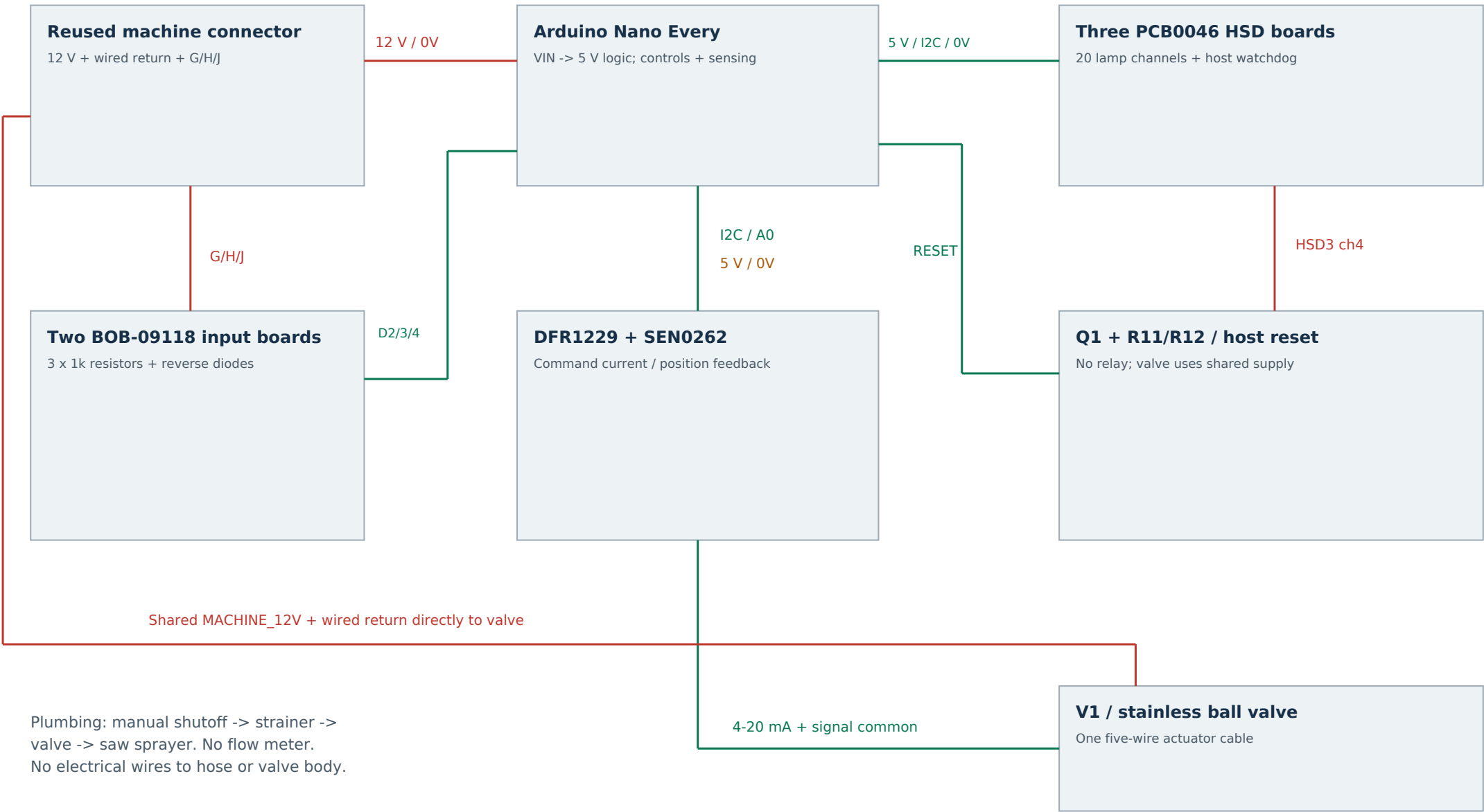
Check	Required evidence
H1 - Machine harness	Identify actual supply, return and G/H/J cavities. X1.1-5 are project terminal names, not Takeuchi connector cavity numbers. Measure key-off behavior.
H2 - Supply and wiring	Existing fuse rating must suit the added wiring. Confirm conductor sizes, voltage under load and shared return connections. No new machine fuse block is specified.
H3 - Purchased parts	Confirm Nano Every, HSD V2, input-board revision, lamp polarity/current and transistor package.
H4 - Valve interface	Confirm five-wire actuator, signal-common compatibility, command compliance, reported-position behavior and startup/shutdown/USB sequencing.
H5 - Firmware and bench	Implement and test internal/external watchdogs, closed startup, faults, settings storage, buttons, current/voltage measurements where fitted, enclosure temperature and water operation.

Changes from Revision E

- Remove K1 (signal relay), U4 (dedicated coil regulator), D4 (coil diode), and C4/C5 (coil regulator capacitors).
- Connect valve RED directly to the same machine supply as Nano VIN and HSD VIN. GREEN connects directly to U2 OUT. YELLOW connects directly to U3 I+.
- Retain both watchdog capabilities without an additional board. HSD3 CH4 now drives a small transistor that resets the Nano; it no longer powers the valve.
- Add Q1 (onsemi 2N3904BU), R11 (4.7 kohm, 5%, 0.25 W) and R12 (1 kohm, 1%, 0.5 W). R12 shares the input-resistor purchase type.
- All five machine connections, +12 V / 0 V / G / H / J, enter together from the left of the continuous single-page drawing.
- Preserve surviving wire IDs. W086, W095 and W097 change endpoints intentionally. W127-W132 are new reset connections. Removed wire IDs are not reused.

ROCK-SAW WATER CONTROL / REVISED ELECTRICAL OVERVIEW

Detailed terminal circuits: sheets 01-06. Every external wire is listed in connections.csv.



OVERVIEW ONLY: grouped links represent multiple conductors. Use the detailed circuit sheets to assemble.

REV F | 07 OCT 2026 | BENCH DESIGN - H1-H5 HOLD | Named nets connect across sheets

Solid dot = joined wires. Crossing without a dot = no connection. Boxes show functions, not physical footprints.

01 / POWER AND LOCAL BUS

Supply domains, module grounds, I2C addresses and Nano connections

X1 / reused connector

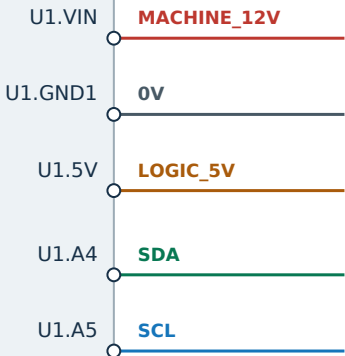
INTERNAL terminal labels only



X1.3 = G; X1.4 = H; X1.5 = J.
Actual machine cavities: H1 HOLD.
No additional fuse block; H2 HOLD.

U1 / Nano Every

Only VIN accepts machine voltage



PARALLEL SUPPLY CONNECTIONS

MACHINE_12V -> TB12 -> U1 VIN + HSD1/2/3 VIN
0V -> TB0 -> all power/signal returns
LOGIC_5V -> TB5 -> U2 VCC, U3 VCC, HSD V,
O1/O2 HV (no flow meter)

HSD load GND also grounds its logic internally.
Do not carry lamp current through a logic header.
U2 OUT_GND = U2 GND internally.
U1 GND2 = U1 GND1 internally.
V1 RED also uses MACHINE_12V; no relay supply.

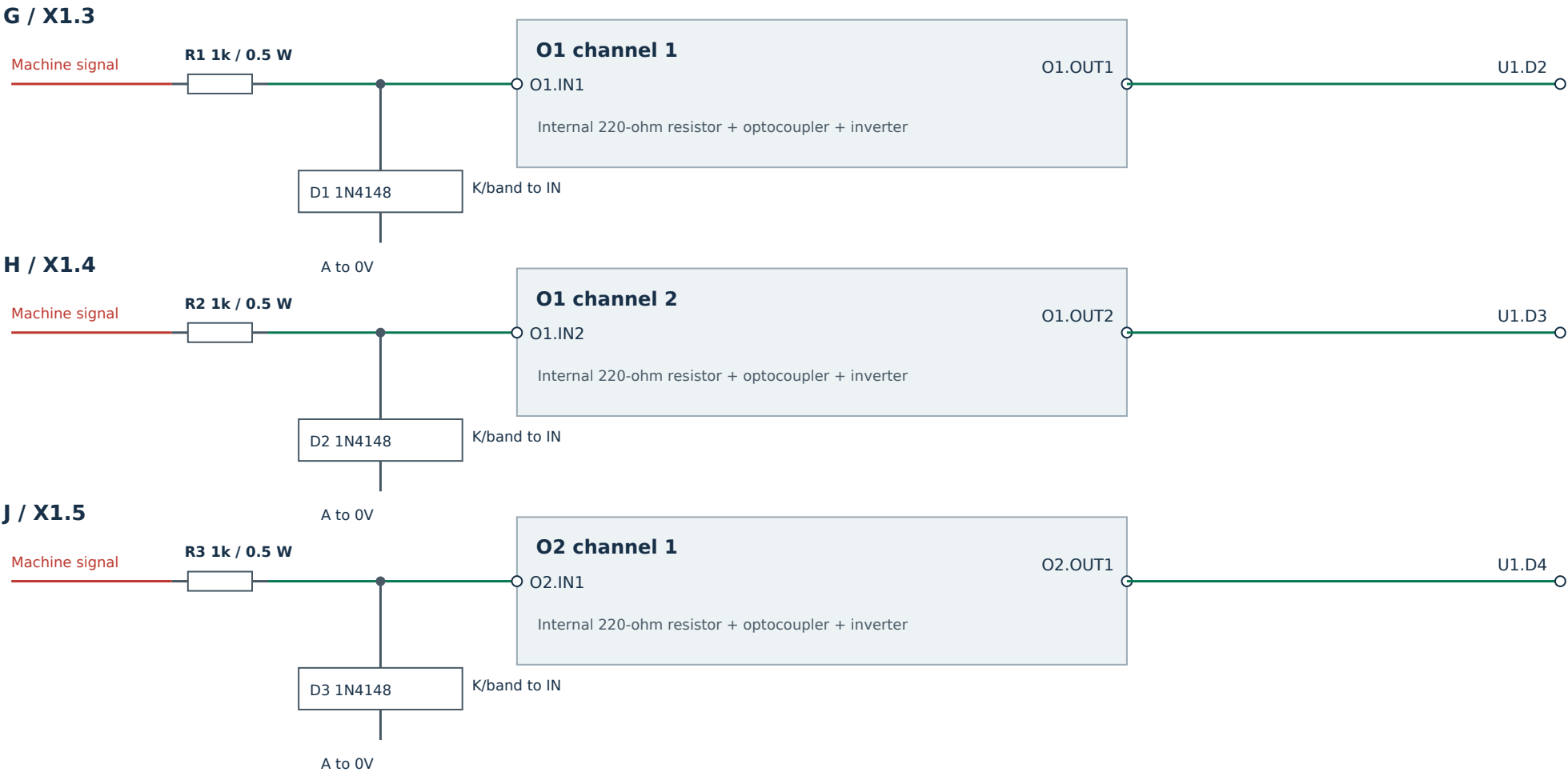
LOCAL I2C - 100 kHz

SDA: U1 A4 -> U2 SDA + HSD1/2/3 D
SCL: U1 A5 -> U2 SCL + HSD1/2/3 C
U2: 0x58; HSD pairs: 0x60/61, 0x62/63, 0x64/65
Use U2 onboard 4.7k pull-ups only.
Open all HSD SDA/SCL pull-up jumpers.
All bus wiring remains inside enclosure.

Local HSD bypass at each VIN/GND: 47 uF / 35 V (+ to VIN) in parallel with 100 nF / 50 V.
USB can energize the logic rail. Disconnect machine, valve and water before USB servicing.

02 / G, H AND J INPUTS

SparkFun BOB-09118 v1.2: HV terminal receives 5 V, not machine voltage



O1/O2 HV -> LOGIC_5V; HV-GND and INPUT_GND -> 0V. O2 IN2 -> INPUT_GND.
O2 OUT2 and both JP2 pin 1 pads unused. HIGH machine input -> HIGH Arduino input.
The three diodes are across the input LED circuits, after the external series resistors.

03 / DIRECT VALVE AND SOLID-STATE WATCHDOG

Shared supply; independent watchdog resets the Arduino, not the valve power

U2 / DFR1229

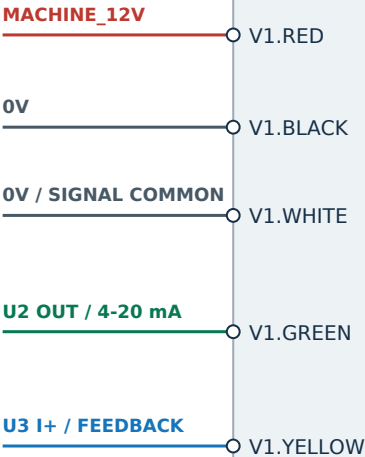
VCC 5 V; GND 0V; SDA A4; SCL A5
OUT -> valve GREEN directly.
OUT_GND = GND -> WHITE / 0V.
Initialize to 4 mA CLOSED before controls.
0 mA is NOT the documented closed command.

U3 / SEN0262

VCC 5 V; GND 0V; SIGNAL -> R4 / A0
I+ -> valve YELLOW directly.
I- -> WHITE / 0V. No added shunt.

V1 / USS-MSV50030

One five-wire actuator cable



HSD3 CH4

R11 4.7k / 5%

Q1.2 / B

R12
1k / 0.5 W

0V

Q1 / 2N3904BU

1 E / 2 B / 3 C

Q1.1 / E 0V

Q1.3 / C U1 RESET

CH4 LOW normally; HIGH on timeout.
Q1 pulls RESET low; never apply 12 V to RESET.
Proposed: 2 s timeout, 100 ms reset pulse.
Return LOW; no automatic repeating pulse.
Reinitialize the watchdog after every reset.

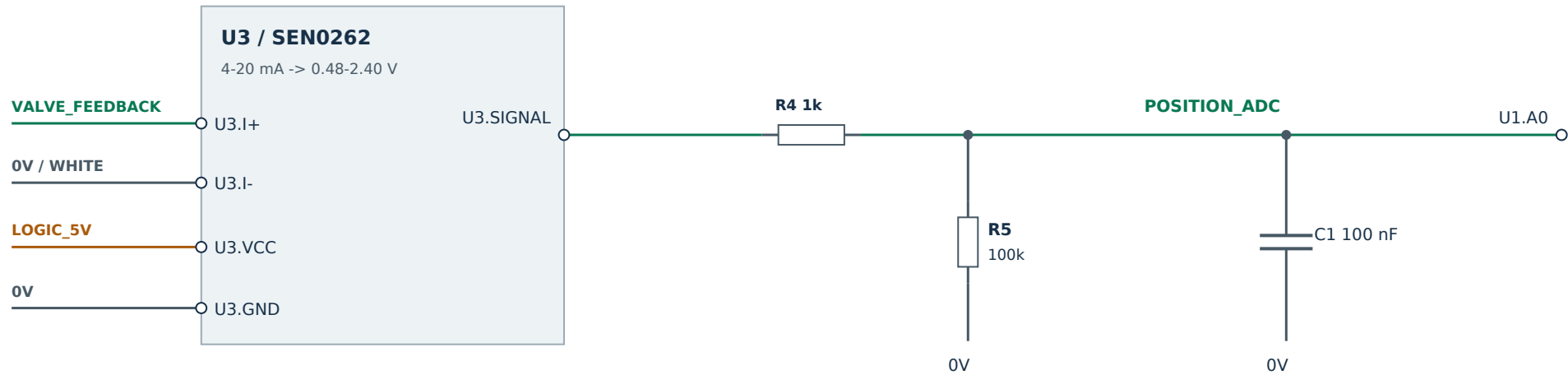
No independent valve-power cut or actuator-current measurement. Reset does not guarantee closure.
Disconnect valve and machine before USB-only service. Verify power sequencing on the bench.

REV F | 07 OCT 2026 | BENCH DESIGN - H1-H5 HOLD | Named nets connect across sheets

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04 / ANALOG POSITION FEEDBACK

No 12 V connection anywhere on this sheet; U2 current-loop output is not an ADC signal



R5 biases an open SIGNAL wire low. R4/R5 reduce the measured voltage by about 1%; calibrate.
No extra 250-ohm resistor across U3 I+/I-. C1 and R5 return locally to the Nano ground.

Revision F omits the flow meter and its entire interface.
Nano D8 and D9 are unused; R6-R9 and C2 are not installed.

Valve position feedback remains required for normal control and recovery.
An indicated closed position is not an independent measurement of stopped water.

Optional manual calibration: collect all nozzle output at known valve positions.
Store a validated monotone flow/position table; there is no automatic meter sweep.

05 / LAMP OUTPUTS AND BOARD CONFIGURATION

Use the printed channel numbers. Each blue/white lamp needs TWO separate switched positives.

HSD1 / PCB0046 V2
0x60/61 | LOAD VIN 12 V; VDD 5 V

CH0

L1 BLUE +

CH1

L1 WHITE +

CH2

L2 BLUE +

CH3

L2 WHITE +

CH4

L3 BLUE +

CH5

L3 WHITE +

CH6

L4 BLUE +

CH7

L4 WHITE +

HSD2 / PCB0046 V2
0x62/63 | LOAD VIN 12 V; VDD 5 V

CH0

L5 BLUE +

CH1

L5 WHITE +

CH2

L6 BLUE +

CH3

L6 WHITE +

CH4

L7 BLUE +

CH5

L7 WHITE +

CH6

L8 BLUE +

CH7

L8 WHITE +

HSD3 / PCB0046 V2
0x64/65 | LOAD VIN 12 V; VDD 5 V

CH0

L9 BLUE +

CH1

L9 WHITE +

CH2

L10 BLUE +

CH3

L10 WHITE +

CH4

HOST RESET / WATCHDOG

CH5

UNUSED / OFF

CH6

UNUSED / OFF

CH7

UNUSED / OFF

VIN -> MACHINE_12V; LOAD GND -> 0V
D -> SDA; C -> SCL; V -> LOGIC_5V
All lamp BLACK leads -> rated 0V return

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All HSD SDA/SCL pull-up jumpers OPEN. Address: HSD1 all open; HSD2 A1 closed; HSD3 A2 closed.
V2 component-side: printed ch7 at top, ch0 at bottom; schematic OUT1 = ch7, OUT8 = ch0.
Top supply: VIN left / G right. Bottom supply: G left / VIN right. Verify markings before power.

06 / TERMINAL ORIENTATION AND ASSEMBLY NOTES

Physical pin references apply ONLY to the stated package and revision

LOW-VOLTAGE CONNECTIONS

Nano D7, D8 and D9 are unused; leave unconnected.
No external temperature probes or flow meter.

Retain valve feedback through U3 to Nano A0.
Position feedback does not measure actual water flow.

C1: at A0 / logic return.
Q1 / R11 / R12: solid-state host reset.
C7-C12: local HSD load-supply bypass.
Removed: FM1, R6-R10, C2, C3, C6, TS1, TS2.
Removed reference numbers are not reused.

POWER AND DIAGNOSTICS

Valve and electronics share MACHINE_12V.
No relay or separate coil regulator.
All load returns go to rated ground distribution.
Keep lamp current out of the Nano/ADC return.

Driver thermal protection remains built in.
Generic driver FAULT does not measure ambient heat.
THER is a control input, not a temperature output.
Codes 2 and 9: no corresponding hardware sensors.

X1 labels are NOT verified machine pin cavities.

Q1 / onsemi 2N3904BU TO-92

Use the exact manufacturer package drawing.
Pin 1 = emitter / 0V; pin 2 = base / R11-R12 node.
Pin 3 = collector / Nano RESET.
Do not substitute a transistor by shape alone.
No added reset pull-up: Nano has 100k to 5 V.

WATCHDOG CONFIGURATION

HSD3 CH4 is reserved; exclude it from lamp writes.
Normal LOW; timeout HIGH; return LOW after 100 ms.
Do NOT use 0xFFFF / permanent HIGH: reset would lock.
Verify both configuration packets and actual SW8B image.
Nano internal watchdog: proposed 1.024 s.
External pulse: proposed 2 s timeout; healthy feed 100 ms.

Component roles

Reference	Component	Function
U1	Genuine Arduino Nano Every ABX00028	Main controller, onboard 5 V regulator and internal hardware watchdog
U2	DFRobot DFR1229 / GP8600	Converts digital commands to valve control current
U3	DFRobot SEN0262	Converts reported valve position current to voltage for A0
HSD1-3	Serial Wombat PCB0046 HSD V2	Twenty lamp outputs; HSD3 CH4 is a host-reset output
O1/O2	SparkFun BOB-09118 v1.2	Three 12 V input channels converted to active-high 5 V logic
V1	U.S. Solid USS-MSV50030	1/2-inch stainless five-wire proportional valve
L1-10	Nilight TL-248BW or verified equivalent	Common negative, separate BLUE and WHITE positive inputs
Q1	onsemi 2N3904BU TO-92	Open-collector pull-down on Nano RESET

Power and local wiring

The reused machine connector supplies MACHINE_12V and 0V. TB12 feeds U1 VIN, HSD1/2/3 VIN and V1 RED in parallel. TB0 collects lamp, actuator and electronic returns. Keep load current out of Nano headers and signal-ground wiring. HSD logic G, load grounds and channel-return terminals are internally common.

The Nano onboard switching regulator supplies TB5 and the logic rails of U2, U3, all three HSDs, and O1/O2 HV. HV on the SparkFun input board means its logic-side voltage: use 5 V, not 12 V. Do not add a second 5 V source. Verify total load and regulator temperature in the enclosure.

Retain one 47 uF / 35 V / 105 C electrolytic and one 100 nF / 50 V X7R ceramic across each HSD VIN and LOAD_GND: C7/C10, C8/C11, C9/C12. Electrolytic positive goes to VIN. Place close to each board with short leads. These are local supply buffers, not comprehensive surge protection.

The 9-16 V envelope used in calculations is a proposed bench range, not a fully qualified supply range. V1 requires at least 9 V at its leads. Existing upstream fuse protection and wiring must be checked together.

G/H/J inputs and main-power-off behavior

X1.3 is G, X1.4 is H and X1.5 is J in this project only. Each enters a 1 kohm / 0.5 W series resistor (R1-R3), followed by a BOB-09118 input and reverse diode (D1-D3). Diode cathode/band faces the board input; anode connects to its INPUT_GND. O1 OUT1/OUT2 connect to Nano D2/D3; O2 OUT1 connects to D4. O2 IN2 is grounded and OUT2 is unused.

The selected board includes 220 ohms in its input LED path. At 12 V, estimated asserted-input draw is $(12 - 1.2) / 1220 = 8.9 \text{ mA}$. About 0.21 Ah is consumed by one continuously asserted input over 24 hours. A brief button press uses very little charge; the machine's own circuit consumption is additional.

With main supply absent, ground retained and USB unplugged, G/H/J energize only the optocoupler input side. Its output pull-ups use controller 5 V, so the specified circuit has no conductive positive-input path into the logic supply. Confirm this with off-state rail measurements. Shared grounds mean this is not complete galvanic isolation of the assembly.

At power-up, request closed, run the one-second-white / one-second-blue lamp test without accepting gestures, then require released J and centered G/H for 100 ms. Discard inputs held through startup. After fault acknowledgement, held G/H must not prevent recovery; consume the old rocker action and require a fresh J press.

Valve command and feedback

Valve lead	Revision F connection
RED	TB12 MACHINE_12V directly
BLACK	TB0 power return
GREEN	U2 OUT directly
WHITE	TB0 signal common, U2 OUT_GND and U3 I-; verify common compatibility
YELLOW	U3 I+ directly

Initialize U2 to its 0-20 mA range, then command 4 mA for closed. The module is not intrinsically restricted to 4-20 mA: code zero produces 0 mA, which the valve manual does not define as closed. Nominal 16-bit codes are 13107 (closed), 39321 (50% travel), and 65535 (full travel). These are valve-position percentages, not calibrated flow percentages.

U2 contains a boosted output stage; never connect OUT to an Arduino input. Test it first with a separate removable 250-ohm precision load, then the actual valve. Its required input burden voltage and loss-of-signal behavior require measurement; matching the words 4-20 mA is not enough to establish the complete interface.

U3 uses a nominal 120-ohm conversion: 4-20 mA gives 0.48-2.40 V. SIGNAL connects through $R4 = 1\text{ kohm}$ to A0. $R5 = 100\text{ kohm}$ and $C1 = 100\text{ nF}$ go from A0 to local Nano ground. The resistor divider gives a nominal 100/101 attenuation: approximately 0.475-2.376 V at A0. Calibrate with the measured 5 V reference. No additional parallel shunt is installed. This input is not galvanically isolated and its resistor/filter does not make it 12 V tolerant.

Valve-reported position is not measured flow or proof that water has stopped. Bucket/stopwatch calibration through the actual nozzles can provide a monotone position/flow table. No automatic meter sweep or live GPM is provided.

Shared-power and USB limitation

Removing the relay removes physical isolation of the signal positives. Shared machine power avoids intentionally leaving the command module energized while switching off the valve alone. It does not prove harmless behavior during brownouts, unplugged valve power, or unequal rail decay. Measure these states during H4.

USB can independently energize the Nano 5 V rail and U2/U3. Before USB-only programming or testing, unplug the machine connector and disconnect the valve cable; isolate the water. Do not connect USB with the unpowered valve's signal leads still attached. No transistor in the new reset circuit isolates the command or feedback loops.

Two watchdogs, no extra board

The Nano ATmega4809 internal hardware watchdog is the first recovery mechanism; approximately 1.024 seconds is the proposed timeout. Feed it only after useful, bounded control-loop progress. A handled fault may be a valid control-loop outcome; do not intentionally reboot continuously just because an acknowledged hardware fault remains. Bound I2C operations and logging.

The existing SW8B at HSD3 address 0x64 provides the second watchdog. CH4 is LOW normally and HIGH on timeout. CH4 connects through $R11 = 4.7\text{ kohm}$ to Q1 base (pin 2). $R12 = 1\text{ kohm}$ connects base to emitter (pin 1 / ground). Collector (pin 3) connects only to Nano RESET. The Nano already has a reset pull-up and capacitor; no external pull-up is added. Never connect CH4's 12 V output directly to RESET.

Use exact onsemi 2N3904BU pin numbering: 1 emitter, 2 base, 3 collector. Verify the manufacturer package view; do not infer pins from an unmarked transistor's shape. Q1 pulls RESET low without sourcing voltage into it. R12 prevents a floating base and shunts small HSD off-state leakage. At 16 V the 4.7 kohm resistor

dissipates about 0.05 W while asserted, within its 0.25 W rating; only a short pulse is intended. Verify off-state leakage and reset voltage on the assembled boards.

Proposed external timing: 2000 ms timeout, check-in every 100 ms, HIGH pulse 100 ms, then return LOW. This is one recovery pulse per arm, not a permanently asserted reset. Reconfigure the watchdog after restart. The currently reviewed firmware decrements its return counter and does not restore that delay merely by refreshing the main countdown: reinitialization is required after a timeout. Confirm behavior in the installed SW8B firmware image.

Use checked protocol writes for both configuration packets: configure channel mode 7 with normal LOW, timeout HIGH and no Wombat self-reset; then explicitly set the return interval to 100 ms. Do not use the library's default false/self-reset setting without overriding its 0xFFFF permanent timeout state: that would hold the Nano in reset. Do not use the unfinished all-pins timeout bitfield. Verify packet failures, incomplete initialization and late check-ins on hardware. The initial default 10 ms interval must also release reset if the second packet fails.

Read the Nano reset cause before clearing it. Internal watchdog reset and external reset should both lead to a closed command and no restored ON/Flush state. An external reset flag alone does not distinguish Q1 from the physical reset button. Send the closed command before nonessential initialization and animations. Lamps and DAC may retain previous outputs until explicitly reinitialized; an MCU reset does not power-cycle them.

HSD3 CH4 must never appear in lamp PWM writes, sweeps or bulk board initialization during operation. Initialize all outputs inactive before assigning the watchdog. Leave CH5-CH7 off. The reset circuit is only a few milliamps; qualify or disable open-load diagnostics for CH4 so its intentionally small load does not cause a false lamp/driver fault. The circuit recovers a host, not a guaranteed shutoff: a jam, dead controller, stuck bus or failed DAC can defeat closure. A watchdog cannot make this valve spring-return on power loss.

Faults and detection limits

Revision F has no independently switchable actuator supply and no actuator-current sensor. Do not claim to detect actuator overcurrent with a lamp-driver board whose outputs no longer supply that actuator. Code 2 has no fitted hardware detector; code 9 remains reserved because there are no external temperature probes. Wombat current and thermal diagnostics apply to its lamp/reset outputs only.

Hardware firmware must use best-effort closed commands when a fault occurs and the command path remains usable, then check reported position within a bounded deadline. If the command or feedback path is broken, report closure as unknown. Do not claim that an inhibit flag stopped the valve motor or water: there is no independent power cut. A persistent jam or uncertain closure requires the upstream manual water shutoff and repair. Fault acknowledgement must not automatically restore water ON or Flush.

The existing browser simulator retains its tested gesture and fault-display scenarios, including simulated faults without fitted detectors. Its movement-inhibit model is not a validated physical response for this relay-free design. Update that hardware-specific model with the eventual firmware; the simulator is not Arduino firmware or evidence of valve shutdown. Keep this limitation visible rather than silently claiming identical fault coverage.

Logging, enclosure and service

Plan a bounded RAM history and redundant EEPROM settings/last-reset records. No logging firmware is supplied. Do not repeatedly write EEPROM while a button is held or a fault persists. Validate records after interrupted power and reject invalid values. Never store an ON state for automatic recovery.

Prefer a gasketed enclosure, sealed glands, supported wiring and removable modules. Conformal coating can protect clean boards while leaving connectors/service points accessible. The Wombat MOSFET outputs reduce load-switching dissipation compared with Darlington drivers; whole-board temperature still needs checking. Do not claim a potting rating for these assembled boards. Permanent resin conflicts with inexpensive replacement. No additional circuit board is required for Q1/R11/R12; mount them on the existing assembly carrier.

Practical bench sequence

1. Inspect every wire and polarity with power absent. Confirm no 12 V connection to 5 V, A0, RESET or G/H/J logic pins. Check transistor pinout and capacitor polarity.
2. Power the Nano and logic modules with valve disconnected. Measure the 5 V rail and module load. Check USB-only service separation.
3. Exercise all three machine inputs, including asserted inputs with main supply off, then power restoration while held.
4. Test one lamp color, then the complete white/blue startup test and animations. Check current, temperature, return wiring and that CH4 is excluded from lamp writes.
5. Test Q1 separately: verify CH4 LOW releases RESET and HIGH asserts it, then a brief pulse releases it. Confirm neither position powers RESET from 12 V. Test one-shot timeout and both watchdog resets with no valve or water connected.
6. Test U2 on the removable load and U3 with known currents. Then verify actual valve wiring, common returns, command range and reported position without pressurized water.
7. Test shared-power sequencing, broken power/signal connections and bounded communication failures. Confirm boot always commands closed; identify any condition requiring manual water shutoff.
8. Run water through the actual nozzles. Verify settings, Max Cap and Flush behavior, calibration, manual shutoff, enclosure heat and recovery after interruptions. Record actual evidence in Validation.

Source register

Reviewed 7 October 2026. Source inspection is not physical qualification.

Source	Used for
Arduino Nano Every schematic	VIN regulator, shared RESET headers, reset pull-up/capacitor, USB power
ATmega4808/4809 datasheet	Internal watchdog, reset flags and timing
Serial Wombat PCB0046	Channels, addresses and diagnostic limits
Watchdog API	Host reset use and timeout configuration
Watchdog firmware source	Return-delay behavior; verify actual SW8B image
onsemi 2N3904 datasheet	Q1 pin numbering and electrical limits
TI TPS4H160-Q1	HSD output leakage, protected MOSFET switching
SparkFun input schematic	Input resistor/LED/inverter topology
U.S. Solid valve manual	Five-wire connections and 4-20 mA position command
DFRobot DFR1229	Current command configuration
DFRobot SEN0262	Position-current conversion

Complete external connection schedule

Every row is a conductor or local soldered connection. Repeated distribution terminals denote joined bus terminals, not a bundle forced into one screw. Module-internal common connections are listed separately. Wire gauges and protection remain H2.

Wire	Net	From	To
W001	MACHINE_12V	TB12.rail	X1.1
W002	MACHINE_12V	TB12.rail	U1.VIN
W003	MACHINE_12V	TB12.rail	HSD1.VIN
W004	MACHINE_12V	TB12.rail	HSD2.VIN
W005	MACHINE_12V	TB12.rail	HSD3.VIN
W006	0V	TB0.rail	X1.2
W007	0V	TB0.rail	U1.GND1
W008	0V	TB0.rail	U2.GND
W009	0V	TB0.rail	U3.GND
W010	0V	TB0.rail	HSD1.LOAD_GND
W011	0V	TB0.rail	HSD2.LOAD_GND
W012	0V	TB0.rail	HSD3.LOAD_GND
W013	0V	TB0.rail	O1.INPUT_GND
W014	0V	TB0.rail	O1.HV_GND
W015	0V	TB0.rail	O2.INPUT_GND
W016	0V	TB0.rail	O2.HV_GND
W017	0V	TB0.rail	V1.BLACK
W018	0V	TB0.rail	V1.WHITE
W019	0V	TB0.rail	U3.I-
W023	LOGIC_5V	TB5.rail	U1.5V
W024	LOGIC_5V	TB5.rail	U2.VCC
W025	LOGIC_5V	TB5.rail	U3.VCC
W026	LOGIC_5V	TB5.rail	HSD1.V
W027	LOGIC_5V	TB5.rail	HSD2.V
W028	LOGIC_5V	TB5.rail	HSD3.V
W029	LOGIC_5V	TB5.rail	O1.HV
W030	LOGIC_5V	TB5.rail	O2.HV
W032	SDA	U1.A4	U2.SDA

Wire	Net	From	To
W033	SDA	U1.A4	HSD1.D
W034	SDA	U1.A4	HSD2.D
W035	SDA	U1.A4	HSD3.D
W036	SCL	U1.A5	U2.SCL
W037	SCL	U1.A5	HSD1.C
W038	SCL	U1.A5	HSD2.C
W039	SCL	U1.A5	HSD3.C
W040	MACHINE_G	X1.3	R1.1
W041	OPTO_G	R1.2	O1.IN1
W042	OPTO_G	R1.2	D1.K
W043	0V	O1.INPUT_GND	D1.A
W044	G_LOGIC	O1.OUT1	U1.D2
W045	MACHINE_H	X1.4	R2.1
W046	OPTO_H	R2.2	O1.IN2
W047	OPTO_H	R2.2	D2.K
W048	0V	O1.INPUT_GND	D2.A
W049	H_LOGIC	O1.OUT2	U1.D3
W050	MACHINE_J	X1.5	R3.1
W051	OPTO_J	R3.2	O2.IN1
W052	OPTO_J	R3.2	D3.K
W053	0V	O2.INPUT_GND	D3.A
W054	J_LOGIC	O2.OUT1	U1.D4
W055	0V	O2.INPUT_GND	O2.IN2
W056	L1_BLUE	HSD1.CH0	L1.BLUE
W057	L1_WHITE	HSD1.CH1	L1.WHITE
W058	0V	TB0.rail	L1.BLACK
W059	L2_BLUE	HSD1.CH2	L2.BLUE
W060	L2_WHITE	HSD1.CH3	L2.WHITE
W061	0V	TB0.rail	L2.BLACK
W062	L3_BLUE	HSD1.CH4	L3.BLUE
W063	L3_WHITE	HSD1.CH5	L3.WHITE
W064	0V	TB0.rail	L3.BLACK

Wire	Net	From	To
W065	L4_BLUE	HSD1.CH6	L4.BLUE
W066	L4_WHITE	HSD1.CH7	L4.WHITE
W067	0V	TB0.rail	L4.BLACK
W068	L5_BLUE	HSD2.CH0	L5.BLUE
W069	L5_WHITE	HSD2.CH1	L5.WHITE
W070	0V	TB0.rail	L5.BLACK
W071	L6_BLUE	HSD2.CH2	L6.BLUE
W072	L6_WHITE	HSD2.CH3	L6.WHITE
W073	0V	TB0.rail	L6.BLACK
W074	L7_BLUE	HSD2.CH4	L7.BLUE
W075	L7_WHITE	HSD2.CH5	L7.WHITE
W076	0V	TB0.rail	L7.BLACK
W077	L8_BLUE	HSD2.CH6	L8.BLUE
W078	L8_WHITE	HSD2.CH7	L8.WHITE
W079	0V	TB0.rail	L8.BLACK
W080	L9_BLUE	HSD3.CH0	L9.BLUE
W081	L9_WHITE	HSD3.CH1	L9.WHITE
W082	0V	TB0.rail	L9.BLACK
W083	L10_BLUE	HSD3.CH2	L10.BLUE
W084	L10_WHITE	HSD3.CH3	L10.WHITE
W085	0V	TB0.rail	L10.BLACK
W086	MACHINE_12V	TB12.rail	V1.RED
W095	VALVE_COMMAND	U2.OUT	V1.GREEN
W097	VALVE_FEEDBACK	V1.YELLOW	U3.I+
W099	POSITION_RAW	U3.SIGNAL	R4.1
W100	POSITION_ADC	U1.A0	R4.2
W101	POSITION_ADC	U1.A0	R5.1
W102	POSITION_ADC	U1.A0	C1.1
W103	0V	U1.GND1	R5.2
W104	0V	U1.GND1	C1.2
W115	MACHINE_12V	HSD1.VIN	C7.+
W116	MACHINE_12V	HSD1.VIN	C10.1

Wire	Net	From	To
W117	0V	HSD1.LOAD_GND	C7.-
W118	0V	HSD1.LOAD_GND	C10.2
W119	MACHINE_12V	HSD2.VIN	C8.+
W120	MACHINE_12V	HSD2.VIN	C11.1
W121	0V	HSD2.LOAD_GND	C8.-
W122	0V	HSD2.LOAD_GND	C11.2
W123	MACHINE_12V	HSD3.VIN	C9.+
W124	MACHINE_12V	HSD3.VIN	C12.1
W125	0V	HSD3.LOAD_GND	C9.-
W126	0V	HSD3.LOAD_GND	C12.2
W127	WATCHDOG_12V	HSD3.CH4	R11.1
W128	RESET_BASE	R11.2	Q1.2
W129	RESET_BASE	Q1.2	R12.1
W130	0V	TB0.rail	Q1.1
W131	0V	Q1.1	R12.2
W132	RESET_N	Q1.3	U1.RESET1

Internal connections - no added wire

HSD1.LOAD_GND = HSD1.G; HSD1.LOAD_GND = HSD1.G0; HSD1.LOAD_GND = HSD1.G1; HSD1.LOAD_GND = HSD1.G2; HSD1.LOAD_GND = HSD1.G3; HSD1.LOAD_GND = HSD1.G4; HSD1.LOAD_GND = HSD1.G5; HSD1.LOAD_GND = HSD1.G6; HSD1.LOAD_GND = HSD1.G7; HSD2.LOAD_GND = HSD2.G; HSD2.LOAD_GND = HSD2.G0; HSD2.LOAD_GND = HSD2.G1; HSD2.LOAD_GND = HSD2.G2; HSD2.LOAD_GND = HSD2.G3; HSD2.LOAD_GND = HSD2.G4; HSD2.LOAD_GND = HSD2.G5; HSD2.LOAD_GND = HSD2.G6; HSD2.LOAD_GND = HSD2.G7; HSD3.LOAD_GND = HSD3.G; HSD3.LOAD_GND = HSD3.G0; HSD3.LOAD_GND = HSD3.G1; HSD3.LOAD_GND = HSD3.G2; HSD3.LOAD_GND = HSD3.G3; HSD3.LOAD_GND = HSD3.G4; HSD3.LOAD_GND = HSD3.G5; HSD3.LOAD_GND = HSD3.G6; HSD3.LOAD_GND = HSD3.G7; U1.GND1 = U1.GND2; U2.GND = U2.OUT_GND; U1.RESET1 = U1.RESET2

Unused external terminals

O1.JP2_1; O2.JP2_1; O2.OUT2; HSD3.CH5; HSD3.CH6; HSD3.CH7; U1.3V3; U1.A1; U1.A2; U1.A3; U1.A6; U1.A7; U1.AREF; U1.RX; U1.TX; U1.D5; U1.D6; U1.D7; U1.D10; U1.D11; U1.D12; U1.D13; U1.D8; U1.D9

Complete parts schedule

Values specify the proposed circuit. Distribution hardware, cable sizes, sealed connectors and enclosure remain subject to physical hold points. Test instruments and the removable DAC test resistor are not installed circuit components.

Ref	Specification	Note
X1	Reused machine harness; verify actual cavities	
TB12	12 V distribution	Joined terminals; hardware and wire rating subject to H2
TB5	Logic 5 V distribution	Joined terminals; hardware and wire rating subject to H2
TB0	Ground distribution	Joined terminals; hardware and wire rating subject to H2
U1	Arduino Nano Every ABX00028	
U2	DFRobot DFR1229	
U3	DFRobot SEN0262	
HSD1	Serial Wombat PCB0046 HSD V2	
HSD2	Serial Wombat PCB0046 HSD V2	
HSD3	Serial Wombat PCB0046 HSD V2	
O1	SparkFun BOB-09118 v1.2	
O2	SparkFun BOB-09118 v1.2	
V1	U.S. Solid USS-MSV50030	
L1	Nilight TL-248BW	
L2	Nilight TL-248BW	
L3	Nilight TL-248BW	
L4	Nilight TL-248BW	
L5	Nilight TL-248BW	
L6	Nilight TL-248BW	
L7	Nilight TL-248BW	
L8	Nilight TL-248BW	
L9	Nilight TL-248BW	
L10	Nilight TL-248BW	
R1	1000 ohm 1% 0.5 W	
D1	1N4148, axial	Band = K / cathode
R2	1000 ohm 1% 0.5 W	
D2	1N4148, axial	Band = K / cathode

Ref	Specification	Note
R3	1000 ohm 1% 0.5 W	
D3	1N4148, axial	Band = K / cathode
R4	1000 ohm 1% 0.25 W	
R5	100000 ohm 1% 0.25 W	
C1	100 nF X7R 50 V	
C10	100 nF X7R 50 V	
C11	100 nF X7R 50 V	
C12	100 nF X7R 50 V	
C7	47 uF 35 V 105 C electrolytic	
C8	47 uF 35 V 105 C electrolytic	
C9	47 uF 35 V 105 C electrolytic	
Q1	onsemi 2N3904BU TO-92	1 emitter, 2 base, 3 collector; verify selected onsemi package drawing
R11	4700 ohm 5% 0.25 W	
R12	1000 ohm 1% 0.5 W	